



DOUBLE DUCTED WITH CAMBERED AIRFOIL BLADE PROFILE SERIES NEI2/D



الهيئة المصرية العامة للمواصفات والجودة
Egyptian Organization for Standardization and Quality



APPLICATION

NEI2/D tunnel axial fans are designed for general purpose of ventilation where large air volume should be extracted and also large air thrust thus it could be used for underground car parkings semi opened areas , ventilation of tunnels , smoke extraction in case of fire as a part of smoke extract system . the fans generate a high – speed and high – pressure directed air jet .

SELECTION

the engineering of tunnel ventilation is based on the geophysical and building characteristics , the traffic expectations as well as the provided safety objectives leads to set out the total thrust in newton that must be obtained by the fans . then the tunnel profile and length will make the designer suggest whether using coupled , singular , more or less near the tunnel access and more or less spaced out where all the above parameters beside of the air speed affects the thrust produced and once the required thrust is determined for every fan the choice will depend on it's dimensional , functional , noise level , the total absorbed power and finally the economical benefit of purchasing

the quick selection curves and the working data charts for various models refers to the actual performances got at the laboratory in compliance with the prescribed rules and a very important something to be noted as the experimental data concerning thrust are included between 83% and 105% of the theoretical data calculated by using the following formula :

THRUST = air density x volume x air speed at outlet

TESTING

→ **Aeraulic test**

After conducting a gauged cone for the inlet according to BS848 and also conducting the two silencers single measurements are taken where the operating point will be nearly at zero static pressure . calculation of speed data is based on inlet-outlet section.

→ **Thrust tests**

The measurement is done by mounting the NEI2/D on a testing bench built up with a surface sliding on a very low friction bearings and as a result the thrust could be measured via transducer mounted on the bench where this bench is located in a very large room in order to keep the ambient air speed as low as possible thus avoiding the effect of floor ,wall and ceiling

→ **Sound test**

The sound levels are measured according to BS848 : part 2 standard by using a semi reverberating chamber calibrated by using a reference sound source of known sound power level $L_w(r)$ which is located at 2 meters at least from the walls where it's sound pressure level is measured by using two microphones $L_p(r)$ located at the following positions (at a distance of 2 and 4 meters with a 45° inclination) in order to make sure that the chamber is sufficiently reverberating : the difference between the 2 microphones cannot exceed

3 dB

Once the source is replaced by the NEI2/D the average sound pressure level detected by the primary microphone is measured .

And finally the sound power of NEI2/D $L_w(i)$ is determined by adjusting the sound power level of the source according to the difference between the sound pressure of the NEI2/D and the reference source :

$$L_w(i) = L_w(r) + L_p(i) - L_p(r)$$

→ **Vibrations**

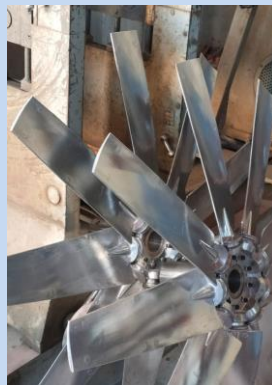
Vibration speed will not exceed 5 mm/s (measurements are taken at the fastening points of the two fan ends complying with ISO DIS 13350

COMPONENTS

→ **HOUSING**

Heavy gauge carbon steel is used with robust flanges at both ends of the fan . the housing is further stiffened in such a way as to allow the fixing of the fan in the tunnel vault . inlet screens are used at both inlet and outlet of the fan . the housing , inlet screens and stiffeners are protected via hot dip galvanized process and also epoxy painting. all bolts used in the process are stainless steel (AISI304).

→ **IMPELLER**



Impellers used are of the airfoil type for best performance and optimum efficiency and also low noise level especially in the high frequency bands

→ **MOTOR**



The used electric motors are suitable for usage in humid , corrosive environment . all are of IP55 Class B or F .

The terminal box incorporates IP55 protection and is mounted externally to the fan housing to help in electrical connections

Both the motor and the terminal box will comply with IEC34-5 and their positioning will comply with IEC34-1

L10 life of motor bearings of 20000 hours at least according to ISO 281 .

FOR EMERGENCY 400 ° for 2 hrs. class H will be used or class F with the motor out of air stream

COMPONENTS

→ CONSTRUCTION

The impeller will be fitted with wing blade profile adjustable pitch angle die cast aluminum blades

Four motor supports will be fastened to the casing through continuous welding process and the unit will be hot dip galvanized according to ISO1459,1460,1461

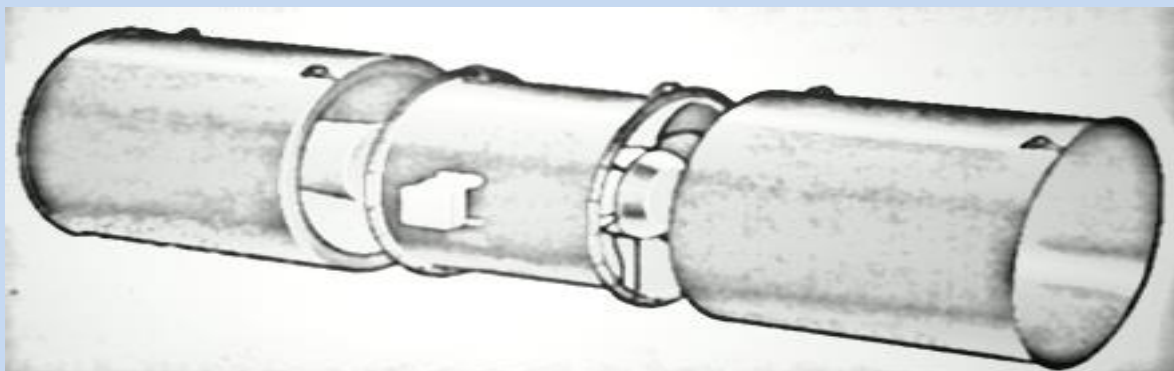
The silencer external casing will be of pre-galvanized steel and epoxy painted

OPTIONAL COMPONENTS

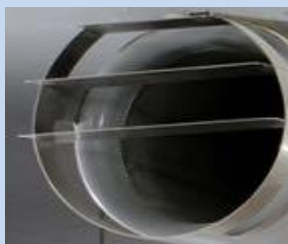
SILENCERS

NEI2/D could be equipped with cylindrical silencers fixed to the housing .

The sound absorbent material used is a special high density rockwool with unperishable and fire proof properties .



GUIDE VANES



Guide vanes could be mounted on the fan discharge or fan intake which guides the flow and improves it's efficiency as it minimizes the total pressure loss

TERMINAL BOX



TECHNICAL CHARACTERISTICS ARE :

1- Wheel size (mm) 500:1250

2- Volume flow (m³/hr.) 4290:175000

3- Thrusts (Newton) 23:2100

→The following performance table data refers to a maximum blade angle for every motor model.

→The sound values in LP are at 3 meters in far field with 45° directivity .

→To know the approximate sound level inside the tunnel add 10 dB

→To the Lp value .



NEI2/D
Performance Diagrams

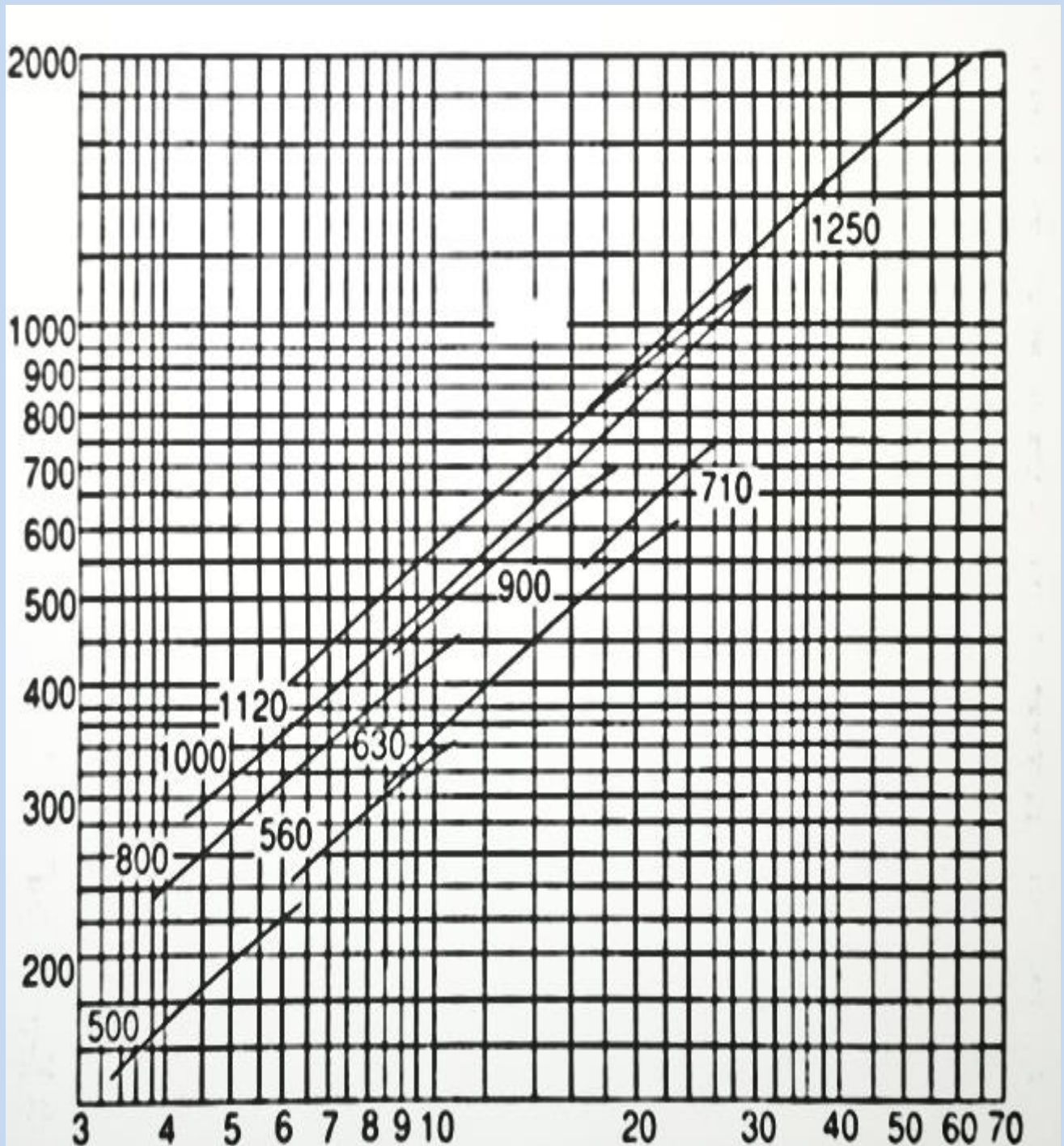


INDUSTRIES

NEI2/D

MODEL	VOLUME FLOW M3/hr	RPM	THRUST N	MOTOR POWER HP	POLE	Sound pressure level LP
NEI2/D 310	4290	2850	23	1	4/2	63
	2107	1400		0.25		
NEI2/D 400	9216	2850	61	2	4/2	70
	4527	1400		0.5		
NEI2/D 450	13122	2850	92	3	4/2	73
	6445	1400		1		
NEI2/D 500	18000	2850	146	5.5	2	69
	21600	2850	215	10	2	72
NEI2/D 560	25200	2850	247	10	2	74
	29520	2850	333	15	2	75
NEI2/D 630	31320	2850	295	15	2	75
	35640	2850	380	20	2	78
	38880	2850	445	25	2	80
	43560	2850	552	40	2	82
NEI2/D 710	49320	2850	545	25	2	81
	52920	2850	635	30	2	83
	55800	2850	730	40	2	84
	61200	2850	938	60	2	86

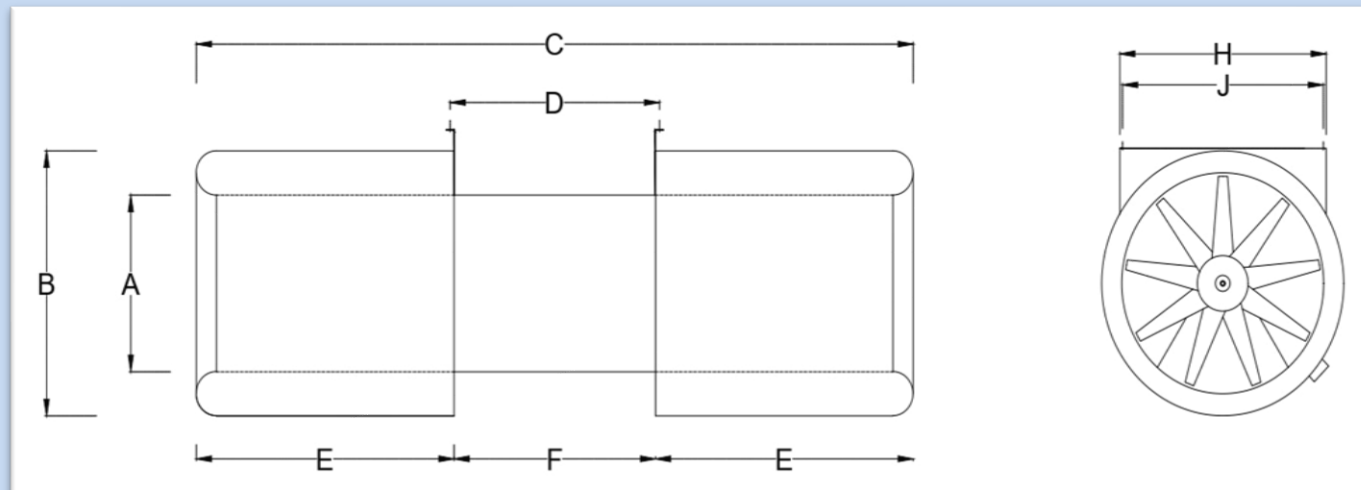
THRUST (N)



ABSORBED POWER (KW)



NEI2/D
Dimensions



MODEL/ DIM. (mm)	A	B	C	D	E	F	G	H	J	KG
NEI2/D310	310	403	1010	340	356.5	297	207	310	279	69
NEI2/D 400	400	520	1304	438	460	384	268	400	360	90
NEI2/D 450	450	585	1467	493	517	432	301	450	405	120
NEI2/D 500	500	650	1630	548	575	480	335	500	450	185
NEI2/D 560	560	710	1890	708	635	620	365	560	510	290
NEI2/D 630	630	780	2030	708	705	620	400	630	580	325
NEI2/D 710	710	860	2410	928	765	840	450	710	660	495
NEI2/D 800	800	1000	2510	796	900	710	510	800	750	420
NEI2/D 900	900	1100	2780	870	1000	780	560	900	850	530
NEI2/D 1000	1000	1200	3090	996	1100	890	610	1000	950	850
NEI2/D 1120	1120	1320	3350	1016	1220	910	710	1120	1070	925
NEI2/D 1250	1250	1450	3610	1016	1350	910	800	1250	1150	1100